

MANUALLY LOADED REPEATER RIFLES

By the 1890s, refinements to the repeater rifle had made it reliable enough for use in many of the world’s armies. In Britain, the .303 Lee-Metford was developed into the Lee-Enfield Mark I of 1895, the first in an enduring line of bolt-action rifles, while in Germany, the Mauser Gewehr 98 of 1898 was a near-faultless design. Thereafter, changes in design were made mostly to reduce the weapon’s weight, or to cut manufacturing costs. After the turn of the century, the barrel length was often reduced to improve handling, although French Berthier and Japanese Arisaka weapons did not follow the general trend.

▼ MAUSER INFANTRIEGEWEHR 98

Date	1898
Origin	Germany
Weight	4.15kg (9lb)
Barrel	74cm (29in)
Calibre	7.92mm × 57

In the Gewehr 98, Mauser brought the bolt action magazine rifle close to perfection by adding a third rear-locking lug, as well as improving gas sealing and refining the magazine. If the rifle had a fault, it lay in the design of its bolt handle.



▲ MOSIN-NAGANT M91

Date	1891
Origin	Imperial Russia
Weight	4.43kg (9¾lb)
Barrel	80.2cm (31½in)
Calibre	7.62mm × 54R

The “3-line”, as it was called, was Imperial Russia’s first repeater rifle and its first in a modern calibre. The “line” was a measure approximating one-tenth of an inch and refers to its calibre.



▲ MANNLICHER M1895

Date	1895
Origin	Austria
Weight	3.78kg (8¼lb)
Barrel	76.5cm (30in)
Calibre	8mm × 50R

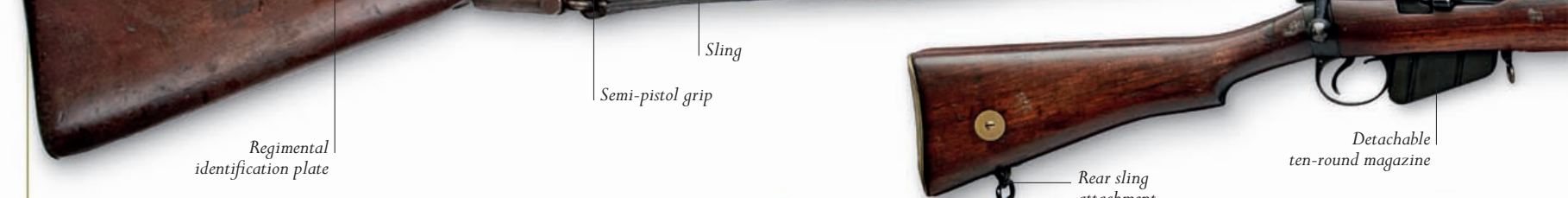
The straight-pull, bolt-action M1895, designed by German engineer Ferdinand von Mannlicher, used a rotating locking lug turned in a camming (spiral) groove. Ammunition was fed from a fixed box magazine.



Bolt handle protrudes horizontally

Rear-locking lug

Leaf-type rear sight



Regimental identification plate

Sling

Semi-pistol grip



Bolt handle

Ten-round magazine

▲ LEE-ENFIELD MARK I

Date	1895
Origin	UK
Weight	4kg (8¾lb)
Barrel	63.5cm (25in)
Calibre	.303in

A redesigned version of the .303 Lee-Metford of 1888, the “.303 calibre, Rifle, Magazine, Lee-Enfield” – or Mark I – had a detachable ten-round magazine, and, with the bolt handle near the trigger, was faster to operate than the rival Mauser rifle.



Wooden butt